

Name :- Aman Sharma

Roll No. :- 2401330130039

Branch :- IT

Section :- A

Workshop Lab Number :- 203D

QUESTION1: Write a Java program to insert multiple records using batch processing.

```
package myapp;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.Statement;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/BatchInsertServlet")

public class BatchInsertServlet extends HttpServlet
{
    private static final long serialVersionUID = 1L;
```

```
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

    response.setContentType("text/html");

    PrintWriter out = response.getWriter();

    String user = "root";
    String pass = "root";

    out.println("<html><head><title>Batch Insert Result</title>"); out.println("</
head><body>");
    out.println("<div>");
    out.println("<h2>Question 1: Batch Insert Execution</h2>");

    try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        // First connect to default DB to ensure advjava exists
        Connection tempCon =
DriverManager.getConnection("jdbc:mysql://localhost:3306/", user, pass);

        tempCon.createStatement().executeUpdate("CREATE DATABASE IF NOT EXISTS
advjava");

        tempCon.close();

        Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/advjava", user, pass);

        // Create table
        Statement stmt = con.createStatement();
```

```
stmt.executeUpdate("CREATE TABLE IF NOT EXISTS q1_students (id INT PRIMARY KEY, name VARCHAR(50), course VARCHAR(50))");
```

```
stmt.executeUpdate("TRUNCATE TABLE q1_students"); // clear for demo
```

```
String insertSql = "INSERT INTO q1_students (id, name, course) VALUES (?, ?, ?)";
```

```
PreparedStatement pstmt = con.prepareStatement(insertSql);
```

```
con.setAutoCommit(false);
```

```
// Record 1
```

```
pstmt.setInt(1, 101);
```

```
pstmt.setString(2, "Alice Smith");
```

```
pstmt.setString(3, "Computer Science");
```

```
pstmt.addBatch();
```

```
// Record 2
```

```
pstmt.setInt(1, 102);
```

```
pstmt.setString(2, "Bob Johnson");
```

```
pstmt.setString(3, "Data Science");
```

```
pstmt.addBatch();
```

```
// Record 3
```

```
pstmt.setInt(1, 103);
```

```
pstmt.setString(2, "Charlie Brown");
```

```
pstmt.setString(3, "Software Engineering");
```

```
pstmt.addBatch();
```

```
int[] results = pstmt.executeBatch();
```

```

con.commit();

out.println("<p>Successfully executed batch insert!</p>");
out.println("<p>Inserted " + results.length + " records into
<code>q1_students</code> table.</p>");

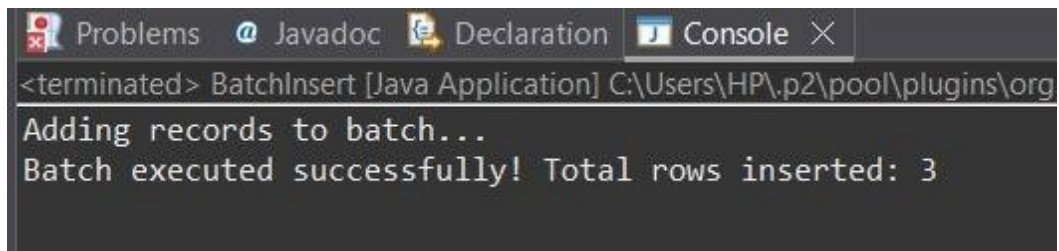
con.close();

} catch (Exception e) {
    out.println("<p>Error: " + e.getMessage() + "</p>");
    e.printStackTrace();
}

out.println("<a href='index.html'>Back to Dashboard</a>"); out.println("</div></
body></html>");
}
}

```

Result Grid			
Filter Rows:			
	id	name	email
▶	1	Alice Smith	alice@example.com
	2	Bob Jones	bob@example.com
	3	Charlie Brown	charlie@example.com
★	NULL	NULL	NULL



```
<terminated> BatchInsert [Java Application] C:\Users\HP\p2\pool\plugins\org
Adding records to batch...
Batch executed successfully! Total rows inserted: 3
```

QUESTION2: Develop a shopping cart application using Session Tracking in Servlets.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>

<%@ page import="java.util.List" %>

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Shopping Cart (Session Tracking)</title>

</head>

<body>

    <div>

        <h2> Online Store</h2>

        <p>Add items to your session-tracked shopping cart:</p>
```

```

<form action="ShoppingCartServlet" method="POST">
  <select name="item" required>
    <option value="Laptop">Laptop - $999</option>
    <option value="Smartphone">Smartphone - $699</option>
    <option value="Headphones">Headphones - $199</option>
    <option value="Keyboard">Mechanical Keyboard - $89</option>
  </select>
  <button type="submit">Add to Cart</button>
</form>

```

```

<div>
  <h3>Your Cart</h3>
  <%
    List<String> cart = (List<String>)
    session.getAttribute("cart"); if (cart == null ||
    cart.isEmpty()) {
      out.println("<p style='color: #94A3B8;'>Your cart is empty.</p>");
    } else {
      for (String item : cart) {
        out.println("<div class='cart-item'>• " + item + "</div>");
      }
      out.println("<br><form action='ShoppingCartServlet' method='POST'><input
type='hidden' name='clear' value='true'><button type='submit' style='background-
color: #EF4444;'>Clear Cart</button></form>");
    }
  %>
</div>

<a href="index.html" class="btn-home">← Back to Dashboard</a>

```

```
</div>
```

```
    </  
body>
```

```
    </  
html>
```

```
package myapp;
```

```
import java.io.IOException;
```

```
import java.util.ArrayList;
```

```
import java.util.List;
```

```
import jakarta.servlet.ServletException;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
```

```
import jakarta.servlet.http.HttpSession;
```

```
@WebServlet("/ShoppingCartServlet")
```

```
public class ShoppingCartServlet extends HttpServlet
```

```
{ private static final long serialVersionUID = 1L;
```

```
@SuppressWarnings("unchecked")
```

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)  
throws ServletException, IOException {
```

```
    HttpSession session = request.getSession(true);
```

```
    List<String> cart = (List<String>) session.getAttribute("cart");
```

```
    if (cart == null) {
```

```
        cart = new ArrayList<>();
```

```
        session.setAttribute("cart", cart);
    }

    String clear = request.getParameter("clear");
    if (clear != null && clear.equals("true")) {
        cart.clear();
    } else {
        String item = request.getParameter("item");
        if (item != null && !item.trim().isEmpty()) {
            cart.add(item);
        }
    }
}

response.sendRedirect("shopping_cart.jsp");
}
```



Online Store

Add items to your session-tracked shopping cart:

Laptop - \$999



Add to Cart

Your Cart

Your cart is empty.

QUESTION3: Write a JSP program using a Declaration tag to declare a variable and display its value.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Question 3: JSP Declaration</title>

</head>

<body>

    <div>

        <h2>JSP Declaration Tag Demo</h2>


        <%!

            // This is a JSP Declaration tag. Variables declared here are instance variables.
            int pageVisitCount = 0;

        %>


        <%

            // This is a scriptlet. We increment the instance variable on every request.
            pageVisitCount++;

        %>


        <p>You are visitor number:</p>

        <div class="counter"><%= pageVisitCount %></div>
```

```
<div class="code-snippet">
```

```
    <lt;%! int pageVisitCount = 0; %Cgt;<br>
```

```
    ...<br>
```

```
    <lt;%= pageVisitCount %Cgt;
```

```
</div>
```

```
<p style="color: #94A3B8; font-size: 0.9rem;">Refresh the page to see the  
declaration variable increment (as it's an instance variable of the generated  
Servlet).</p>
```

```
<a href="index.html" class="btn-home">← Back to Dashboard</a>
```

```
</div>
```

```
</  
body>
```

```
</  
html>
```

QUESTION4: Employee Payroll Management. Scenario: An organization maintains employee records including Employee ID, Name, Department, Salary, and City.

- 1. Insert employee details using JDBC.**
- 2. Display employees whose salary is greater than ₹50,000.**
- 3. Update the salary of a specific employee.**
- 4. Delete an employee record.**
- 5. Display the total number of employees.**

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8"%>

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Employee Payroll Management</title>

</head>

<body>

    <h1>Employee Payroll Management</h1>

    <div>

        <!-- 1. Insert Employee -->

        <div class="card">

            <h3>1. Insert Employee</h3>

            <form action="EmployeeServlet" method="POST">

                <input type="hidden" name="action" value="insert">

                <input type="number" name="empId" placeholder="Employee ID" required>
```

```
<input type="text" name="name" placeholder="Name" required>
<input type="text" name="dept" placeholder="Department" required>
<input type="number" name="salary" placeholder="Salary" required>
<input type="text" name="city" placeholder="City" required>
<button type="submit">Add Employee</button>
</form>
</div>
```

```
<!-- 3. Update Salary -->
<div class="card">
  <h3>3. Update Salary</h3>
  <form action="EmployeeServlet" method="POST">
    <input type="hidden" name="action" value="update">
    <input type="number" name="empId" placeholder="Employee ID to Update"
required>
    <input type="number" name="newSalary" placeholder="New Salary" required>
    <button type="submit">Update Salary</button>
  </form>
</div>
```

```
<!-- 4. Delete Employee -->
<div class="card">
  <h3>4. Delete Employee</h3>
  <form action="EmployeeServlet" method="POST">
    <input type="hidden" name="action" value="delete">
    <input type="number" name="empId" placeholder="Employee ID to Delete"
required>
    <button type="submit" class="danger">Delete Employee</button>
  </form>
```

```
</div>
```

```
<!-- Displays and Results -->
```

```
<div class="card">
```

```
  <h3>Queries C Results</h3>
```

```
  <div style="display: flex; gap: 1rem; margin-bottom: 1rem;">
```

```
    <form action="EmployeeServlet" method="GET" style="flex: 1; margin: 0;">
```

```
      <input type="hidden" name="action" value="displayRich">
```

```
      <button type="submit">2. Display Salary > ₹50,000</button>
```

```
    </form>
```

```
  <form action="EmployeeServlet" method="GET" style="flex: 1; margin: 0;">
```

```
    <input type="hidden" name="action" value="count">
```

```
    <button type="submit" style="background-color: #22C55E;">5. Count Total  
Employees</button>
```

```
  </form>
```

```
</div>
```

```
<div class="result-box">
```

```
  <%
```

```
    String msg = request.getParameter("msg");
```

```
    if (msg != null) {
```

```
      out.println(msg);
```

```
    } else {
```

```
      out.println("Ready. Execute an action to see results here.");
```

```
    }
```

```
  %>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<a href="index.html" class="btn-home">← Back to Dashboard</a>
```

```
</body>
```

```
</html>
```

```
package myapp;
```

```
import java.io.IOException;
```

```
import java.sql.Connection;
```

```
import java.sql.DriverManager;
```

```
import java.sql.PreparedStatement;
```

```
import java.sql.ResultSet;
```

```
import java.sql.Statement;
```

```
import java.net.URLEncoder;
```

```
import jakarta.servlet.ServletException;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
```

```
@WebServlet("/EmployeeServlet")
```

```
public class EmployeeServlet extends HttpServlet {
```

```
    private static final long serialVersionUID = 1L;
```

```
    private static final String URL = "jdbc:mysql://localhost:3306/";
```

```

private static final String DB_URL = "jdbc:mysql://localhost:3306/advjava";
private static final String USER = "root";
private static final String PASS = "root";

private Connection getConnection() throws Exception
{
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection tempCon = DriverManager.getConnection(URL, USER, PASS);
    tempCon.createStatement().executeUpdate("CREATE DATABASE IF NOT EXISTS
advjava");
    tempCon.close();

    Connection con = DriverManager.getConnection(DB_URL, USER, PASS);
    Statement stmt = con.createStatement();
    stmt.executeUpdate("CREATE TABLE IF NOT EXISTS Employee (empId INT PRIMARY
KEY, name VARCHAR(100), dept VARCHAR(50), salary DOUBLE, city VARCHAR(50))");
    return con;
}

protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
    String action = request.getParameter("action");
    String msg = "";

    try (Connection con = getConnection())
    {
        if ("insert".equals(action)) {
            String sql = "INSERT INTO Employee (empId, name, dept, salary, city) VALUES (?,
?, ?, ?, ?)";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setInt(1, Integer.parseInt(request.getParameter("empId")));

```

```

        ps.setString(2, request.getParameter("name"));
        ps.setString(3, request.getParameter("dept"));
        ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));
        ps.setString(5, request.getParameter("city"));
        ps.executeUpdate();
        msg = "Successfully inserted employee record.";
    }
    else if ("update".equals(action)) {
        String sql = "UPDATE Employee SET salary = ? WHERE empId = ?";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setDouble(1, Double.parseDouble(request.getParameter("newSalary")));
        ps.setInt(2, Integer.parseInt(request.getParameter("empId")));
        int rows = ps.executeUpdate();
        msg = rows > 0 ? "Successfully updated salary." : "Employee ID not found.";
    }
    else if ("delete".equals(action)) {
        String sql = "DELETE FROM Employee WHERE empId = ?";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setInt(1, Integer.parseInt(request.getParameter("empId")));
        int rows = ps.executeUpdate();
        msg = rows > 0 ? "Successfully deleted employee record." : "Employee ID not found.";
    }
} catch (Exception e) {
    msg = "Error: " + e.getMessage();
}

response.sendRedirect("employee.jsp?msg=" + URLEncoder.encode(msg,
"UTF-8"));

```



```
}
```

```
protected void doGet(HttpServletRequest request, HttpServletResponse response)  
throws ServletException, IOException {
```

```
    String action = request.getParameter("action");
```

```
    StringBuilder msg = new StringBuilder();
```

```
    try (Connection con = getConnection())
```

```
    { if ("displayRich".equals(action)) {
```

```
        Statement stmt = con.createStatement();
```

```
        ResultSet rs = stmt.executeQuery("SELECT * FROM Employee WHERE salary >  
50000");
```

```
        msg.append("Employees earning > ₹50,000:\n\n");
```

```
        boolean found = false;
```

```
        while (rs.next())
```

```
        { found = true;
```

```
            msg.append("ID: ").append(rs.getInt("empId"))
```

```
                .append(" | Name: ").append(rs.getString("name"))
```

```
                .append(" | Dept: ").append(rs.getString("dept"))
```

```
                .append(" | Salary: ₹").append(rs.getDouble("salary"))
```

```
                .append(" | City: ").append(rs.getString("city")).append("\n");
```

```
        }
```

```
        if (!found) msg.append("No employees found matching criteria.");
```

```
    }
```

```
    else if ("count".equals(action)) {
```

```
        Statement stmt = con.createStatement();
```

```
        ResultSet rs = stmt.executeQuery("SELECT COUNT(*) AS total FROM  
Employee");
```

```
        if (rs.next()) {
```

```

        msg.append("Total number of employees: ").append(rs.getInt("total"));
    }
}
} catch (Exception e) {
    msg.append("Error: ").append(e.getMessage());
}

response.sendRedirect("employee.jsp?msg=" + URLEncoder.encode(msg.toString(),
"UTF-8"));
}
}

```



Employee Payroll Management

1. Insert Employee

Employee ID Name Department Salary City

3. Update Salary

Employee ID to Update New Salary

4. Delete Employee

Employee ID to Delete

Queries & Results

Ready. Execute an action to see results here.